EXTENSION BY ZERO IS COMPATIBLE WITH PUSHING FORWARD ALONG A COMPACTIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G4X (sources/stacks-duality/pinned-original-source.tex line 8577).

Allowed standard prerequisites (editorial).

- The notion of a compactification over the base. More on Flatness, Section labelled flat-section-compactify, cited at proof lines 8612-8616. Fixes the meaning of the hypothesis.
- The saturated case of the statement. Same chapter, Lemma labelled lemma-well-defined-pre, cited at proof lines 8618-8620. Settles the case j'(U) = f^{-1}(j(U)).
- Cofilteredness of compactifications and its part (c). More on Flatness, Lemma labelled flat-lemma-compactifications-cofiltered part (c), cited at proof lines 8626-8630. Gives the two isomorphisms used in the general case.

Source reference. Original reference 0G4N: sources/stacks-duality/pinned-original-source.tex, source lines 8203–8203. Same-chapter pointer into the author environment remark at source line 8203; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0ATT: sources/stacks-chow/tags.txt, source lines 13626–13626. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8591 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G4W: sources/stacks-duality/pinned-original-source.tex, source lines 8472–8472. Same-chapter pointer into the author environment lemma at source line 8472; invoked at proof line 8593 of the counted proof.

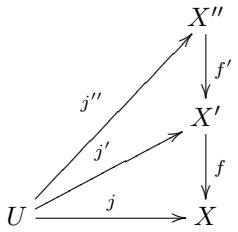
Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ATU: sources/stacks-chow/tags.txt, source lines 13627–13627. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8606 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $j : U \to X$ be an open immersion of Noetherian schemes. Let $j' : U \to X'$ be a compactification of U over X (see proof) and denote $f : X' \to X$ the structure morphism. Then we have a canonical isomorphism $Rj_! \to Rf_* \circ R(j')_!$ of functors $D^b_{Coh}(\mathcal{O}_U) \to \text{Pro-}D^b_{Coh}(\mathcal{O}_X)$ where $Rj_!$ and $Rj'_!$ are as in Remark 0G4N.*

Proof. The fact that X' is a compactification of U over X means precisely that $f : X' \to X$ is proper, that j' is an open immersion, and $j = f \circ j'$. See More on Flatness, Section 0ATT. If $j'(U) = f^{-1}(j(U))$, then the lemma follows immediately from Lemma 0G4W. If $j'(U) \neq f^{-1}(j(U))$, then denote $X'' \subset X'$ the scheme theoretic closure of $j' : U \to X'$ and denote $j'' : U \to X''$ the corresponding open immersion. Picture



By More on Flatness, Lemma 0ATU part (c) and the discussion above we have isomorphisms $Rf'_* \circ Rj''_! = Rj'_!$ and $R(f \circ f')_* \circ Rj''_! = Rj_!$. Since $R(f \circ f')_* = Rf_* \circ Rf'_*$ we conclude. □